

KARTIK DWIVEDI

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Experience

[Hypergro.ai](#)

Software Engineer (Intern)

Jun 2026 – Present

Bangalore - Remote

- Built an end-to-end **AI video generation platform** (editor, avatar, storyboard) orchestrating **6+ Google Vertex AI/Gemini models** via a **config-driven tool-calling pipeline** processing thousands of daily jobs.
- Architected a **fault-tolerant async job queue** using **BullMQ and GCP Cloud Tasks**; integrated parallel fan-out execution, **idempotency locks**, and isolated failure handling to achieve **scale-to-zero infrastructure**.
- Implemented **enterprise RBAC & OAuth authentication** across 40+ authorization checkpoints, remediating **multi-tenancy IDOR** vulnerabilities and establishing per-user AI spend guardrails.

[GuildUp](#)

Backend Engineer Intern

May 2025 – Aug 2025

Remote

- Designed and shipped a **fault-tolerant, event-driven automation system (fan-out architecture)** powering real-time reminders, post-session nudges, and feedback workflows across the platform.
- Built a **unified coupon and automation engine** with behavioral triggers, and streamlined **CI/CD workflows** on GCP for reliable, repeatable deploys.
- Owned **end-to-end** feature delivery from requirements to deployment and monitoring and cut backend latency using **asynchronous job queues**.

Projects

[Ditto – Duplicate-Logic Detector](#)

[Live](#) | [Demo](#) | June 2026 - July 2026

- Architected a **dynamic AST analysis engine** using **ts-morph and Node.js VM** to detect functional logic duplicates; scanned **5,800+ open-source functions** across 5 repositories and isolated **163 duplicate logic clusters**.
- **Engineered sandboxed execution** routines to validate function equivalence via **real runtime outputs**, detecting a critical silent bug in the **cline repository (64k+ starred repository)** missed by static analysis tools.
- Optimized **LLM token consumption** by designing a function-level chunking strategy with **fingerprint based clustering**, cutting analysis costs to **\$2 per codebase** while maintaining output accuracy. Every new PR run costs less than \$0.1 per run.
- Deployed production **web interface on GCP Cloud Run** with automated Docker CI/CD pipelines, serving real-time requests for active external users.
- **Tech Stack:** TypeScript, Node.js, Express, MongoDB, OpenAI (embeddings + structured outputs), sandboxed execution (worker_threads / vm), ts-morph, GCP Cloud Run, Docker, Next.js.

[BlueStellers – Expert Consultation & Booking Marketplace](#)

Oct 2025 - June 2026

- Engineered a dual-app consultation marketplace ([Expert Dashboard](#) & [Client Platform](#)) using Next.js, Node.js, and MongoDB across 2 independently deployed **microservices** with **automated Google Calendar/Meet integration**.
- Integrated **Razorpay payment** infrastructure handling **webhooks, automated refunds, and coupon application logic**; implemented **failure-recovery workflows** to guarantee **zero-loss transaction state reconciliation**.
- Hardened **application security** and reliability using **cross-domain authentication, AES calendar token encryption at rest, and an async background-job pipeline** on GCP Cloud Tasks & Redis BullMQ
- **Tech Stack:** Next.js, TypeScript, Node.js/Express, MongoDB, Redis + BullMQ, Razorpay, Google Calendar/Meet, JWT/next-auth, Firebase, WhatsApp API, GCP (Cloud Run, Cloud Tasks), Vercel

Technical Skills

Languages: Java, JavaScript, TypeScript, Python, SQL

Frameworks & Web: Node.js, Express.js, Next.js, React.js, FastAPI

Databases & Queues: PostgreSQL, MySQL, MongoDB, Redis, Kafka, RabbitMQ

Cloud & DevOps: GCP, AWS, Docker, CI/CD, GitHub Actions, Git, Jest

Architecture & AI: Distributed Systems, System Design, Microservices, RAG, Embeddings

Achievements

- Finalist, **Smart India Hackathon (SIH) 2024** – AI-powered face authentication with deepfake detection.
- **Runner-up, Hitachi & GlobalLogic 30Hacks** – ML-powered transport optimization app; won **2 Lakhs cash prize**.
- Solved **900+ algorithmic problems** on LeetCode & codeforces. ([Profile](#))
- Authored multiple technical blogs on **HashNode**. ([Profile](#))

Education

National Institute of Technology, Raipur

Bachelor of Technology - Mechanical Engineering

2021 – 2025

CPI: 8.13/10